

Premeasure and The Extension Theorem The workflow (12) is already very promising except one key issue:

Carathéodary Family can be Small

What if $\mathcal{M}_{\mu^*}$ is trivial or very small?

The next theorem prevents this by starting with an *algebra* $\mathcal{A}$ instead of just a plain family of sets $\mathcal{E}$. The plain notion of size on $\mathcal{E}$ is replaced by the so-called *premeasure*.

Definition 1.3 (Premeasure)

Suppose $\mathcal{A} \subset \mathcal{P}(X)$ is an algebra, a function $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ is a **premeasure** if

1. $\mu_0(\emptyset) = 0$
2. Let $\{A_j\}_1^\infty$ is a sequence of disjoint sets in $\mathcal{A}$. Given $\bigcup_1^\infty A_j \in \mathcal{A}$ then

$$\mu_0\left(\bigcup_1^\infty A_j\right) = \sum_1^\infty \mu_0(A_j) \quad (18)$$

Exercise 1.2 (Completeness of $\mathcal{M}_{\mu^*}$)

A measure μ on a σ -algebra $\mathcal{M}$ is complete if

$$\forall N \in \mathcal{M} \quad \forall N' \subseteq N \quad (\mu(N) = 0 \Rightarrow N' \in \mathcal{M}). \quad (19)$$

Let μ^* be an outer measure on a set X . Prove that the measure on $\mathcal{M}_{\mu^*}$ induced by μ^* by the Carathéodory construction is complete. (The proof of Carathéodory theorem in the book does not explain this well, write a detailed proof.)

Solution to Exercise 1.1

Proof 1.1

Note that $\mu(N) = 0 \Rightarrow \mu(N') = 0$. $\forall E \subseteq X$, by subadditivity:

$$\mu^*(E) \leq \mu^*(E \cap N') + \mu^*(E \cap N'^c) \quad (20)$$

By monotonicity and positivity of μ^* we have $\mu^*(E \cap N') = 0$ (since $E \cap N' \subseteq N'$ so $\mu^*(E \cap N') \leq 0$). But again by monotonicity $\mu^*(E \cap N'^c) \leq \mu^*(E)$ hence we have:

$$\mu^*(E) \leq \mu^*(E \cap N') + \mu^*(E \cap N'^c) \leq \mu^*(E) \quad (21)$$

Hence:

$$\mu^*(E) = \mu^*(E \cap N') + \mu^*(E \cap N'^c) \quad (22)$$

As desired.

Exercise 1.3

Let $\mathcal{A} \subset \mathcal{P}(X)$ be an algebra, $\mathcal{A}_\sigma$ the collection of countable unions of sets in $\mathcal{A}$, and $\mathcal{A}_{\sigma\delta}$ the collection of countable intersections of sets in $\mathcal{A}_\sigma$. Let μ_0 be a premeasure on $\mathcal{A}$ and μ^* the induced outer measure.

1. For any $E \subset X$ and $\epsilon > 0$ there exists $A \in \mathcal{A}_\sigma$ with $E \subset A$ and $\mu^*(A) \leq \mu^*(E) + \epsilon$.

2. If $\mu^*(E) < \infty$, then E is μ^* -measurable iff there exists $B \in \mathcal{A}_{\sigma\delta}$ with $E \subset B$ and $\mu^*(B \setminus E) = 0$.
3. If μ_0 is σ -finite, the restriction $\mu^*(E) < \infty$ in (2) is superfluous.

Solution to Exercise 1.1

1. This is just the definition of outer measure (namely, infimum). Note since

$$\mu^*(E) = \inf C \quad (23)$$

where

$$C = \left\{ \sum \mu_0(A_j) \mid A = \cup A_j \in \mathcal{A}_\sigma, E \subset A \right\} \quad (24)$$

Hence $\exists c \in C$ such that $c < \mu^*(E) + \epsilon$ by the definition of infimum. Now the map between open covers and their area is clearly surjective by construction:

$$\begin{aligned} \{A \in \mathcal{A}_\sigma : E \subset A\} &\rightarrow \mathbb{R} \\ A &\mapsto \underbrace{\sum_j \mu_0(A_j)}_{\mu^*(A)} \end{aligned} \quad (25)$$

hence we can find $A \in \mathcal{A}_\sigma$ that corresponds to c .

3. ($\implies$) Suppose E is μ^* -measurable. Now by (1), for each n we can find $A_n \in \mathcal{A}_\sigma$ with $E \subset A_n$ and $\mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$. The A_n need not be decreasing, so replace them by $\tilde{A}_n := \bigcap_{k=1}^n A_k$; since $\mathcal{A}_\sigma$ is closed under finite intersections (intersection distributes over the countable unions defining each A_k), $\tilde{A}_n \in \mathcal{A}_\sigma$, and now $\tilde{A}_n$ is a decreasing sequence with $E \subset \tilde{A}_n \subset A_n$, so still $\mu^*(E) \leq \mu^*(\tilde{A}_n) \leq \mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$. Define:

$$B := \bigcap_n \tilde{A}_n = \bigcap_n A_n \in \mathcal{A}_{\sigma\delta} \quad (26)$$

By construction $E \subset B$. Since $\tilde{A}_n \in \mathcal{A}_\sigma \subset \mathcal{M}_{\mu^*}$ is a decreasing sequence with $\mu^*(\tilde{A}_1) \leq \mu^*(E) + 1 < \infty$, *continuity-from-above* on $\mathcal{M}_{\mu^*}$ gives:

$$\mu^*(B) = \lim_{n \rightarrow \infty} \mu^*(\tilde{A}_n) = \mu^*(E) \quad (27)$$

Now use that E is μ^* -measurable, testing it against B : since $E \subset B$,

$$\mu^*(B) = \mu^*(B \cap E) + \mu^*(B \cap E^c) = \mu^*(E) + \mu^*(B \setminus E) \quad (28)$$

Combining with $\mu^*(B) = \mu^*(E) < \infty$ and subtracting the finite common value $\mu^*(E)$ from both sides gives $\mu^*(B \setminus E) = 0$, as desired.

($\impliedby$) Suppose there exists $B \in \mathcal{A}_{\sigma\delta}$ such that $E \subset B$ and $\mu^*(B \setminus E) = 0$. Note any $B \in \mathcal{A}_{\sigma\delta}$ is μ^* -measurable since $\mathcal{A} \subset \mathcal{M}_{\mu^*}$ (by *extension theorem*). We would like for all $F \subset X$

$$\mu^*(F) \geq \mu^*(F \cap E) + \mu^*(F \cap E^c) \quad (29)$$

But now $E \subset B$ hence:

$$\begin{aligned} \mu^*(F) &= \mu^*(F \cap B) + \mu^*(F \cap B^c) \\ &\geq \mu^*(F \cap E) + \mu^*(F \cap B^c) \end{aligned} \quad (30)$$

Note all we need now is $\mu^*(F \cap B^c) \geq \mu^*(F \cap E^c)$.

But now

$$\mu^*(F \cap E^c) = \mu^*(F \cap (B^c \cup (B \cap E^c))) \leq \mu^*(F \cap B^c) + \mu^*(F \cap (B \cap E^c)) \quad (31)$$

Again, by monotonicity and the assumption

$$\mu^*(F \cap (B \cap E^c)) \leq \mu^*(B \cap E^c) = 0 \quad (32)$$

Therefore $\mu^*(F \cap B^c) \geq \mu^*(F \cap E^c)$. As desired.

4. Suppose μ_0 is σ -finite then $\exists \{X_n\}$, $\mu_0(X_n) < \infty$ such that $X = \cup_n X_n$. Now define $E_n = X_n \cap E$, we can apply (2) to each E_n (There existss $B_n \in \mathcal{A}_{\sigma\delta}$ satisfying the property iff $E_n \in \mathcal{M}$).

($\implies$) Say we already have desired set B_n for each E_n (each $B_n \in \mathcal{A}_{\sigma\delta}$, $E_n \subset B_n$, $\mu^*(B_n \setminus E_n) = 0$)^a. Take the union $B = \cup_n B_n$; it covers E since $E_n \subset B_n$ and $E = \cup_n E_n$. It also satisfies the null-excess property: since $E_n \subset E$ we have $B_n \setminus E \subset B_n \setminus E_n$ for each n , so $B \setminus E = \cup_n (B_n \setminus E) \subset \cup_n (B_n \setminus E_n)$, and by countable subadditivity:

$$\mu^*(B \setminus E) \leq \sum_n \mu^*(B_n \setminus E_n) = \sum_n 0 = 0 \quad (33)$$

($\impliedby$) Suppose there is such B for E (possibly of infinite outer-measure). Define $E_n = X_n \cap E$ as before, and now also $B_n := X_n \cap B \in \mathcal{A}_{\sigma\delta}$ (intersecting with $X_n \in \mathcal{A}$ preserves membership in $\mathcal{A}_{\sigma\delta}$). Then $E_n \subset B_n$, and

$$\mu^*(B_n \setminus E_n) = \mu^*(X_n \cap (B \setminus E)) \leq \mu^*(B \setminus E) = 0 \quad (34)$$

Since $\mu^*(E_n) \leq \mu_0(X_n) < \infty$, part (2) applies to E_n and gives $E_n \in \mathcal{M}$. Now we want to show $E \in \mathcal{M}$ but this is trivial as $\mathcal{M}$ is a σ -algebra closed under countable union and $E = \cup_n E_n$

^aStrictly, $\mathcal{A}_{\sigma\delta}$ need not be closed under *countable* unions (only finite ones) — the analogous fact for G_δ sets in $\mathbb{R}$ fails: $\mathbb{Q} = \bigcup_q \{q\}$ is a countable union of G_δ singletons yet is not itself G_δ . So $B = \cup_n B_n$ below should be read as living in $\mathcal{M}_{\mu^*}$ (a genuine σ -algebra), which is all that is actually used afterward.

Exercise 1.4

Let μ^* be an outer measure on X induced from a finite premeasure μ_0 . If $E \subset X$, define the inner measure of E to be $\mu_*(E) = \mu_0(X) - \mu^*(E^c)$. Then E is μ^* -measurable iff $\mu^*(E) = \mu_*(E)$

Solution to Exercise 1.1

($\implies$) Suppose E is μ^* -measurable. Then so is E^c , and since $X = E \sqcup E^c \in \mathcal{A}$, additivity gives $\mu_0(X) = \mu^*(X) = \mu^*(E) + \mu^*(E^c)$, i.e. $\mu^*(E) = \mu_*(E)$.

($\impliedby$) Suppose $\mu^*(E) + \mu^*(E^c) = \mu_0(X) = \mu^*(X) < \infty$. Fix $\epsilon > 0$; by Exercise 1.1 part (1), pick $A, A' \in \mathcal{A}_\sigma \subset \mathcal{M}_{\mu^*}$ with $E \subset A$, $E^c \subset A'$, $\mu^*(A) \leq \mu^*(E) + \epsilon$, $\mu^*(A') \leq \mu^*(E^c) + \epsilon$. Then $A \cup A' = X$, so

$$\mu^*(X) + \mu^*(A \cap A') = \mu^*(A) + \mu^*(A') \leq \mu^*(X) + 2\epsilon \implies \mu^*(A \cap A') \leq 2\epsilon \quad (35)$$

Since $A \setminus E \subset A \cap A'$, $\mu^*(A \setminus E) \leq 2\epsilon$. Taking $\epsilon = \frac{1}{n}$ gives $A_n \supset E$ with $\mu^*(A_n \setminus E) \leq \frac{2}{n}$; let $B := \bigcap_n A_n \in \mathcal{M}_{\mu^*}$. Then $E \subset B$ and $\mu^*(B \setminus E) \leq \mu^*(A_n \setminus E) \rightarrow 0$, so $\mu^*(B \setminus E) = 0$.

This is exactly the hypothesis of the ($\impliedby$) direction of Exercise 1.1 part (2) (only measurability of B was used there, not $B \in \mathcal{A}_{\sigma\delta}$), which concludes E is μ^* -measurable. As desired.

The following exercise is assumed everywhere in mathematics and is not directly related to outer measure. But they are invoked multiple times, for instance, in subadditivity of outer measure.

Exercise 1.5 (Order invariance for Series of Positive Terms)

1. Prove that any generalized series with summands in $[0, \infty]$ can be summed in any order. Specifically, if $a_j \in [0, \infty]$ for $j \in \mathbb{N} = \{1, 2, 3, \dots\}$, then for any bijection $\mathbb{N} \rightarrow \mathbb{N}$, $j \mapsto k_j$, the equality $\sum_{j=1}^{\infty} a_j = \sum_{j=1}^{\infty} a_{k_j}$ holds. (That is, a nonnegative generalized series can be summed in any order.)
2. Prove that any two generalized series with summands in $[0, \infty]$ can be summed as one series. Specifically, if $a_j, b_j \in [0, \infty]$ for $j \in \mathbb{N}$, then the equality $\sum_{j=1}^{\infty} a_j + \sum_{j=1}^{\infty} b_j = \sum_{j=1}^{\infty} (a_j + b_j)$ holds.

Solution to Exercise 1.1

1. We first make a claim

$$\sum_{j \in \mathbb{N}} a_j = \sup_J \sum_{j \in J} a_j \quad (36)$$

where $J \subset \mathbb{N}, |J| < \infty$.

- To show $\sum_{j \in \mathbb{N}} a_j \leq \sup_J \sum_{j \in J} a_j$:

$$\sum_{j \in \mathbb{N}} a_j = \lim_{N \rightarrow \infty} \sum_{j=1}^N a_j \leq \sup_J \sum_{j \in J} a_j \quad (37)$$

- To show $\sum_{j \in \mathbb{N}} a_j \geq \sup_J \sum_{j \in J} a_j$, we first note $\forall J$ as above:

$$\sum_{j \in \mathbb{N}} a_j = \sup_m s_m \geq s_{\max J} \geq \sum_{j \in J} a_j \quad (38)$$

Now sup preserves non-strict inequality so we can take sup on both sides to get:

$$\sup_J \sum_{j \in \mathbb{N}} a_j = \sum_{j \in \mathbb{N}} a_j \geq \sup_J \sum_{j \in J} a_j \quad (39)$$

as desired. Now (1) becomes trivial since $\forall J$ defined as above, suppose we define:

$$k(J) = \{k(j) : j \in J\} \quad (40)$$

now by the claim

$$\sum_{j \in \mathbb{N}} a_{k(j)} = \sup_J \sum_{j \in J'} a_j = \sup_J \sum_{j \in k(J)} a_j \quad (41)$$

where the last equality follows as J' is also in the set $\{J \subset \mathbb{N} : |J| < \infty\}$.

1. We first make a claim: For monotone non-decreasing sequences s_n, t_n :

$$\sup(s_n + t_n) = \sup s_n + \sup t_n \quad (42)$$

- $\sup(s_n + t_n) \leq \sup s_n + \sup t_n$ is true since $\forall m, s_m + t_m \leq \sup s_n + \sup t_n$ and taking $\sup_m$ on both sides get us desired.

- $\sup(s_n + t_n) \geq \sup s_n + \sup t_n$ can be shown by arguing $\forall \epsilon > 0$,

$$\sup(s_n + t_n) \geq \sup s_n + \sup t_n - \epsilon \quad (43)$$

Denote $s = \sup s_n, t = \sup t_n$. By definition of supremum, we can pick N_1, N_2 such that $s_{N_1} > s - \epsilon/2$ and $t_{N_2} > t - \epsilon/2$. Let $N = \max(N_1, N_2)$; since s_n, t_n are non-decreasing, $\forall n \geq N$ we have $s_n \geq s_{N_1} > s - \epsilon/2$ and $t_n \geq t_{N_2} > t - \epsilon/2$, hence:

$$s_n + t_n > s + t - \epsilon \quad (44)$$

Taking $\sup_n$ on the left gives $\sup(s_n + t_n) \geq s + t - \epsilon = \sup s_n + \sup t_n - \epsilon$. As desired, since $\epsilon > 0$ was arbitrary. (2) then easily follows by taking the claim in (1) and then let

$$\begin{aligned} s_n &= \sum_{j=1}^n a_j \\ t_n &= \sum_{j=1}^m b_j \end{aligned} \quad (45)$$

Apply the above claim gets desired.

Casestudy: Lebesgue-Stieltjes Measure

The most useful set that we would like to equip a measure with is, unarguably, the real numbers $\mathbb{R}$.

Starting Point: The half-intervals Recall that we would start with an elementary collection that has a clear notion of size. We define the **h-intervals**:

$$H = \{(a, b] : a < b, a, b \in \mathbb{R}\} \quad (46)$$

We define a terminology here:

Definition 1.4 (Borel Sets)

Let X any metric space, the σ -algebra generated by the family of open sets in X is called the **Borel σ -algebra**, denoted by $\mathcal{B}_X$

Exercise 1.6 (h-intervals generate $\mathcal{B}_{\mathbb{R}}$)

Check H is an elementary set and so it generates an algebra $\mathcal{A}$. Furthermore, show that the σ -algebra generated by $\mathcal{A}$ is $\mathcal{B}_{\mathbb{R}}$

Another terminology:

Definition 1.5 (Elementary Decomposition)

Suppose $A \in \mathcal{A}$, since $\mathcal{A} = \mathcal{M}(H)$ we can find finite disjoint h-intervals $\{I_j\}_{j=1}^N = \{(a_j, b_j]\}_{j=1}^N$ such that:

$$A = \bigcup_{j=1}^N I_j \quad (47)$$

Then $\{I_j\}$ is an **elementary decomposition** of A .

We define size on H as follows: